DATA SCIENCE

# Machine Learning Starter Pack

The algorithms, metrics and concepts that cover most entry-level data science interviews.

- **7 algorithms and when to use each**
- **The 99% accuracy trap**
- **60-day plan from zero**

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# 7 algorithms

You do not need 40. These cover most real tabular data problems.

- **Linear Regression**
  Predict a number. Next month sales.
- **Logistic Regression**
  Predict yes or no. Will this customer churn?
- **Decision Tree**
  Easy to explain to business. Clear if-then rules.
- **Random Forest**
  Many trees voting. Strong first model on most datasets.
- **XGBoost / LightGBM**
  Usually the best accuracy on tabular data.
- **K-Means**
  Group similar customers when there is no label.
- **KNN**
  Predict from the nearest similar records. Simple baseline.

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# Pick the right metric

Accuracy alone can lie. Choose by what a mistake costs.

- **Precision**

  Of everything flagged, how much was right? Use when false alarms are costly, like spam.

- **Recall**

  Of all real cases, how many did we catch? Use when missing one is costly, like fraud or disease.

- **F1 score**

  Balance of precision and recall.

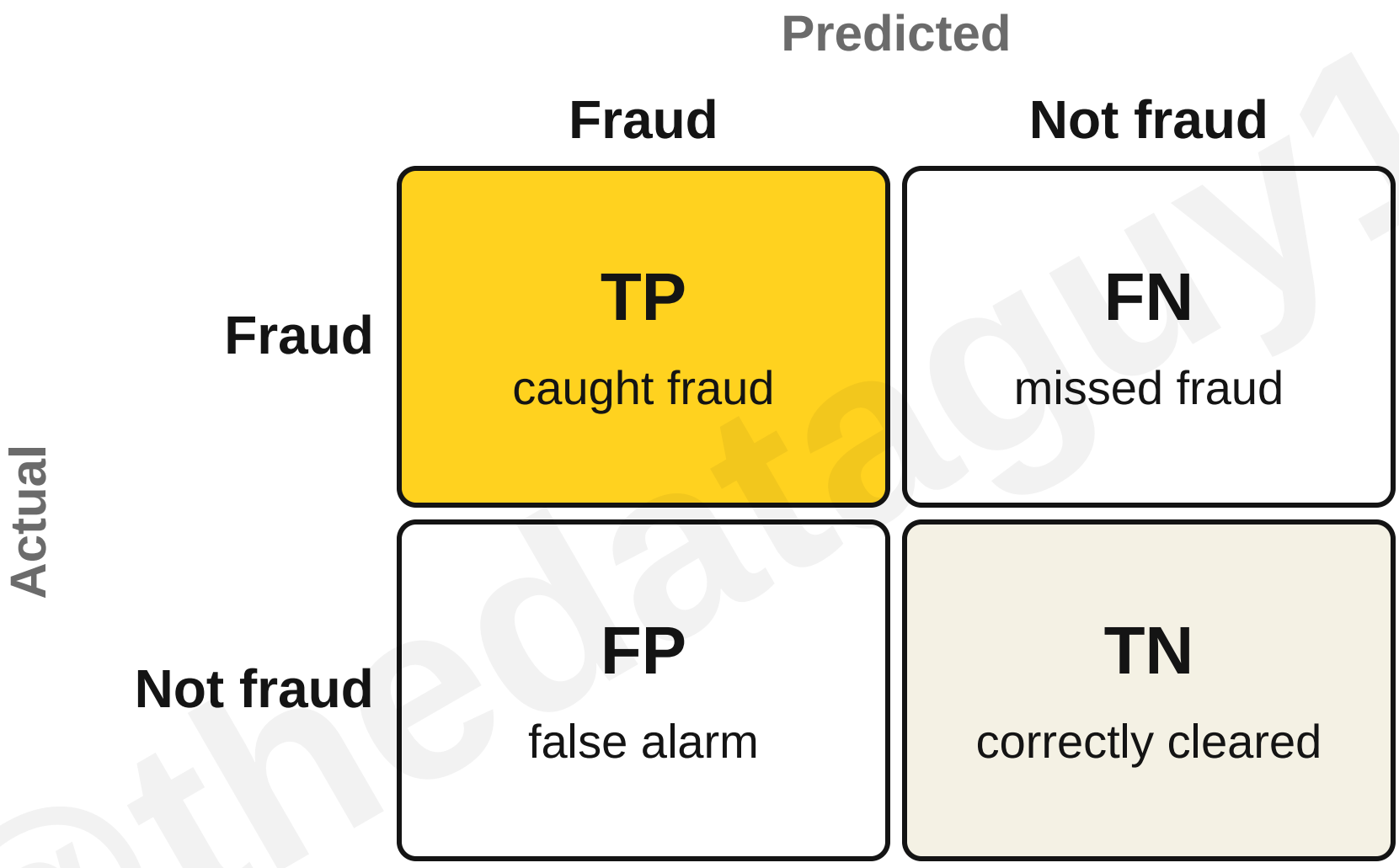
- **RMSE / MAE**

  Average error when predicting a number. MAE is easier to explain.

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# Read a confusion matrix

Every classification metric comes from these 4 boxes.




```
Precision = TP / (TP + FP)
Recall    = TP / (TP + FN)
Accuracy  = (TP + TN) / total
```

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# "99% accurate. Why is that bad?"

A favourite interview question. Here is the full answer.

## The trap

If 1% of transactions are fraud, a model that always says "not fraud" is 99% accurate and catches zero fraud.

## What to check

Confusion matrix, recall and precision on the fraud class.

## How to fix

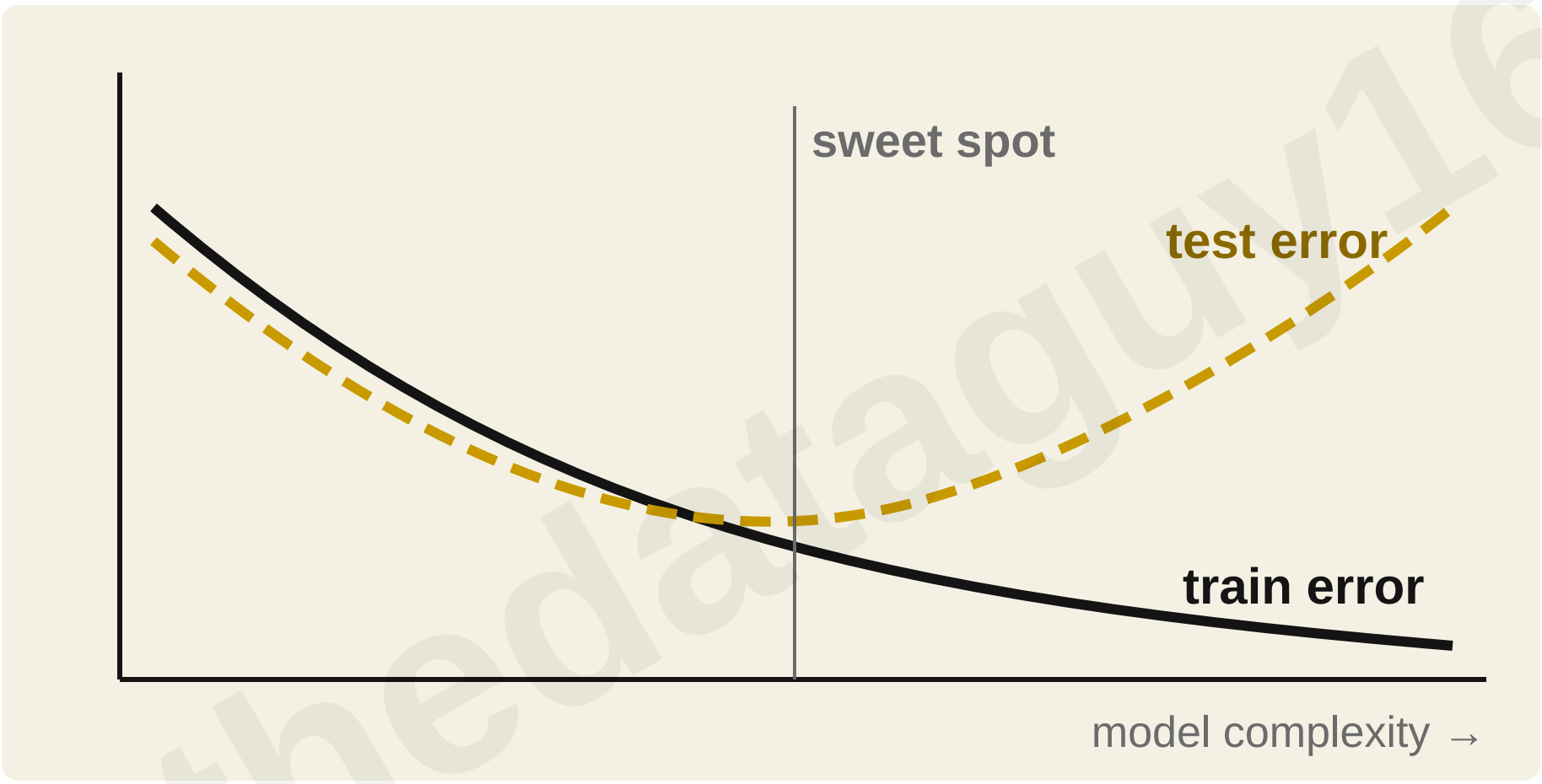
Class weights, oversampling (SMOTE), or lowering the decision threshold.

```
RandomForestClassifier(class_weight="balanced")
```

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# Overfitting in one picture

The model memorised the training data instead of learning the pattern.




## Signs

Train score 98%, test score 70%.

## Fixes

More data, simpler model, regularisation, cross-validation, fewer features.

## Underfitting

Both scores low. Model is too simple.

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# 60 days from zero

Two projects at the end are worth more than five certificates.

- **Days 1-10: Python and Pandas**
  Load, clean, group, merge, plot.
- **Days 11-20: Statistics**
  Distributions, hypothesis testing, regression.
- **Days 21-45: Scikit-learn**
  The 7 algorithms, train-test split, cross-validation, metrics.
- **Days 46-60: Two projects**
  Churn prediction and customer segmentation, each with a business recommendation.

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# ML project workflow

Interviewers ask "walk me through your project". Use these steps.

```
from sklearn.model_selection import
train_test_split
from sklearn.ensemble import
RandomForestClassifier
from sklearn.metrics import
classification_report

X_tr, X_te, y_tr, y_te =
train_test_split(
    X, y, test_size=0.2, stratify=y,
random_state=42)
model =
RandomForestClassifier(class_weight="balanced")
model.fit(X_tr, y_tr)
print(classification_report(y_te,
model.predict(X_te)))
```

## 1. Business problem

What decision will the model support?

## 2. Clean, explore, build features

Most of the time goes here.

## 3. Baseline, explain

Beat a simple baseline, then show top features and the money or time saved.

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